

# LAB 3 – ARCHITECTURE SELECTION & JUSTIFICATION

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## Inter-City Freight Load Matching Marketplace

### 1. Architecture Selection

We chose Microservices Architecture for the Inter-City Freight Load Matching Marketplace.

### 2. Two Reasons for Selection

1. Independent scalability: The Load & Bid, Shipment & POD, and Payment services can be scaled independently because bidding activity and payment processing may have different workloads.

2. Fault isolation: If the bidding or shipment service experiences a failure, the Payment Service can remain isolated, reducing the impact on other marketplace functions.

### 3. Security Advantage

Microservices allow authentication and authorization to be enforced at individual service APIs. Access to sensitive payment and escrow operations can be restricted to the Payment Service, while carrier verification and shipment data can be protected through controlled service access.

### 4. Performance Benefit

Load searching and bidding can be scaled horizontally during periods of high bidding activity without unnecessarily scaling shipment tracking or payment services. This helps reduce performance bottlenecks and improves response time.

### 5. Components & Interfaces

The five components are Load & Bid Service, Order Manager Component, Payment Service Component, Shipment & POD Service, and Data Management Service. The four interfaces are Load/Bid API, Payment API, Shipment/POD API, and Data API.

### 6. Conclusion

Microservices are suitable for this marketplace because bidding, shipment/POD, data, order management, and payment functions can be developed, deployed, scaled, and isolated independently.